Project Title

PROJECT REPORT

Submitted by

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Student 2 Name	CB.EN.U4CYS220XX
Student 3 Name	CB.EN.U4CYS220XX
Student 4 Name	CB.EN.U4CYS220XX

in partial fulfillment for the award of the degree

of

Bachelor of Technology

in

Computer Science And Engineering (Cyber Security)

TIFAC-CORE IN CYBER SECURITY

AMRITA SCHOOL OF COMPUTING

AMRITA VISHWA VIDYAPEETHAM

COIMBATORE - 641 112

NOVEMBER 2025

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Under the guidance of

Dr./Mr. GuideName
Guide Designation
Amrita Vishwa Vidyapeetham
Coimbatore

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BONAFIDE CERTIFICATE

This is to certify that this project entitled **"Project Title"** submitted by **Student 1 Name (CB.EN.U4CYS220XX), Student 2 Name (CB.EN.U4CYS220XX), Student 3 Name (CB.EN.U4CYS220XX), Student 4 Name (CB.EN.U4CYS220XX),** **(), ()** in partial fulfillment of the requirements for the award of the **Degree of Bachelor of Technology** in **Computer Science And Engineering (Cyber Security)** is a bonafide record of the work carried out by them under my guidance and supervision at Amrita School of Computing, Coimbatore.

Dr./Mr. GuideName
Guide Designation

Prof. M. Sethumadhavan
Professor and Head

This project report was evaluated by us on _____.

INTERNAL EXAMINER

EXTERNAL EXAMINER

AMRITA VISHWA VIDYAPEETHAM

AMRITA SCHOOL OF COMPUTING, COIMBATORE

TIFAC-CORE IN CYBER SECURITY

DECLARATION

We hereby declare that this project entitled **"Project Title"** is a record of the original work done by us under the guidance of **Dr./Mr. GuideName**, Guide Designation, TIFAC-CORE in Cyber Security, Amrita Vishwa Vidyapeetham and that this work has not formed the basis for the award of any degree, diploma, associateship, fellowship, or any other similar title, to any candidate in any university, to the best of our knowledge.

Place: Coimbatore **Date:** _____

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COUNTERSIGNED

Prof. M. Sethumadhavan
Professor and Head,
TIFAC-CORE in Cyber Security

Acknowledgment

At the very outset, we would like to offer our heartfelt gratitude to the **Almighty**, whose blessings gave us the strength, wisdom, and perseverance to complete this project.

We express our sincere thanks to our guide, **Dr./Mr. GuideName**, Guide Designation, TIFAC-CORE in Cyber Security, Amrita Vishwa Vidyapeetham, for his/her (mention here).

We are immensely grateful to **Prof. M. Sethumadhavan**, Professor and Head, TIFAC-CORE in Cyber Security, for his (mention here).

We would also like to extend our heartfelt thanks to **Mr. Sitaram Chamarty**, Professor of Practice, TIFAC-CORE in Cyber Security, Amrita Vishwa Vidyapeetham, for his (mention here).

We wish to acknowledge the contributions of our family, friends, and seniors for their constant encouragement, emotional support, and constructive discussions. Their curiosity and questions helped us view our work from new perspectives and strengthened our understanding of the subject.

We are also thankful to all the faculty members of TIFAC-CORE in Cyber Security, whose direct or indirect support and encouragement played a vital role in the successful completion of this project.

Abstract

This project addresses a relevant and timely problem in the field of [**insert domain here**] by proposing a solution that is both innovative and practical. The topic explored in this work has significant applications in real-world scenarios, making it highly relevant for academic and industry needs.

The motivation for choosing this topic came from the growing importance of [**insert problem area or trend**]. Current solutions lack efficiency, scalability, or security, leading to the exploration of alternative approaches.

Through a comprehensive review of the literature, various existing methodologies and frameworks were studied. These works helped identify the gaps and opportunities that this project aims to address.

The primary contribution of this project is the design and implementation of [**insert your system, model, or method**]. The proposed approach offers improvements in [**performance, usability, accuracy, etc.**] and incorporates novel ideas to solve the identified problem.

Experimental analysis or theoretical validation has shown promising results, indicating the effectiveness of the proposed system. The results demonstrate clear advantages over the baseline approaches and open avenues for future enhancements or research extensions.

Please rewrite the entire abstract in your own words.

Keywords: Keyword1, Keyword2, Keyword3, Keyword4, and Keyword5

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List of Symbols

a	–	Acceleration
t	–	Time
Δ	–	Change operator
μ	–	Coefficient of friction

(Remove the above and add the symbols used in this report here manually)

List of Abbreviations

USB – Universal Serial Bus
LED – Light Emitting Diode
SDG – Sustainable Development Goals
(Remove the above and add the abbreviations used in this report here manually)

Chapter 1

Introduction

1.1 Motivation

The motivation behind this work stems from the growing need for [insert motivation: e.g., secure data handling, automation, etc.]. As systems become more interconnected and data-driven, addressing [briefly mentioning a key challenge] is essential.

1.2 Problem Statement

The core problem addressed in this project is [state the problem clearly]. It affects [stakeholders or systems] by [impact description]. This project aims to explore solutions that mitigate or resolve these issues through [approach/technique].

1.3 Threat Model

Fig. 1.1 illustrates the threat modeling for a typical College Management System (CMS), identifying potential vulnerabilities and attack surfaces.

1.4 Goals

The primary goals of this project are:

- Design and implement a secure and efficient [system/tool/method].
- To analyze and mitigate key vulnerabilities in [domain/system].
- To ensure scalability, performance, and usability of the proposed solution.

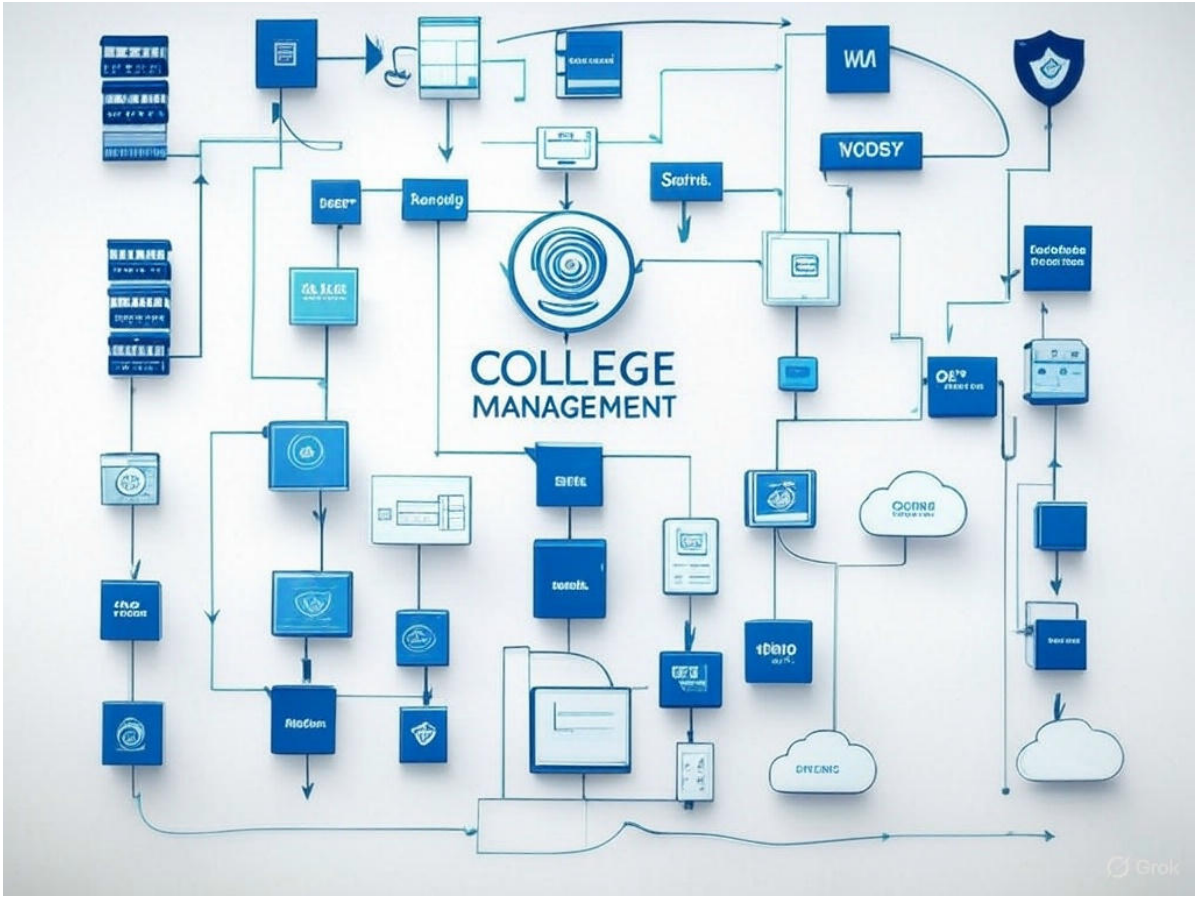

Figure 1.1: Threat Model of College Management System

1.5 Research Methodology

The research is guided by the following questions:

1. What are the current challenges in [domain]?
2. How can [specific method/technique] be used to address these challenges?
3. What are the performance and security implications of the proposed approach?

1.6 Structure of the Project

The remainder of the report is organized into five main chapters. Chapter 2 provides the background and related work necessary to understand the domain, offering insights into existing techniques and systems. Chapter 3 presents a comprehensive literature survey, analyzing the strengths and limitations of previous research, and identifying the gaps that this work aims to address. Chapter 4 details the proposed methods, including architecture, design decisions, implementation steps, and testing methodologies. Chapter 5

discusses the experimental setup, the results obtained, the analysis of the findings, and the comparison with existing methods. Finally, Chapter 6 concludes the report by summarizing the key contributions, highlighting limitations, and suggesting directions for future research.

Chapter 2

Background

This chapter describes [1] work in the literature [1].

Chapter 3

Literature Survey

The literature survey provides information on previous research relevant to the current study. It establishes the foundation, identifies gaps, and helps to justify the significance of the work.

Table 3.1 demonstrates how a table can be included in LaTeX and referenced in the text.

Table 3.1: Record of Student Marks

Roll No	**Name**	**Subject**	**Marks**
101	Alice	Mathematics	87
102	Bob	Physics	78
103	Charlie	Chemistry	92
104	Diana	Biology	85

Chapter 4

Proposed Methods

4.1 Architecture Diagram

This section presents the overall architecture of the proposed system. It includes major modules, their interactions, and data flow between components. [Insert diagram and explanation here.]

4.2 Design Methods

The design phase focuses on selecting appropriate models, defining the data flow, and specifying the behavior of the system. Design considerations include modularity, scalability, and security. [Add relevant diagrams and descriptions.]

4.3 Implementation Methods

This section describes the algorithms and techniques used to implement the proposed solution. An example of this is shown in Algorithm 1, which finds the maximum element in an array.

Algorithm 1 Find Maximum Element in an Array

```
procedure FindMax(A, n)                    ▷ A is the array, n is its size
    max ← A[1]
    for i ← 2 to n do
        if A[i] > max then
            max ← A[i]
        end if
    end for
    return max
end procedure
```

4.4 Testing Methods

Testing ensures that all components function as expected and meet the desired requirements. Unit testing, integration testing, and performance testing are carried out for each module.

As demonstrated, you can also cite a previous chapter—for instance, refer to Chapter 2 for the problem statement and objectives.

Chapter 5

Contribution

5.1 Innovations in the Proposed Method

5.2 Improved Performance and Security Metrics

5.3 Novel Findings or Unexpected Outcomes

Chapter 6

Results and Analysis

6.1 Performance Evaluation

6.2 Strengths & Weaknesses

6.3 Security Analysis

Chapter 7

Conclusion

Chapter 8

Future Work

References

[1] Ramaguru, R., Sindhu, M., Sethumadhavan, M. (2019). Blockchain for the Internet of Vehicles. In: Singh, M., Gupta, P., Tyagi, V., Flusser, J., Ören, T., Kashyap, R. (eds) Advances in Computing and Data Sciences. ICACDS 2019. Communications in Computer and Information Science, vol 1045. Springer, Singapore. https://doi.org/10.1007/978-981-13-9939-8_37

Please use APA format for citing your references.

Appendix A

Source Code

A.1 Server

```python
def hello():
    print("Hello, World!")
```

Listing A.1: Python: Hello World Function

Appendix B

Screenshot

Plagiarism Report

This section includes the plagiarism reports generated for the B.Tech project document. As per academic requirements, two separate reports are provided:

- **With Reference:** The report includes all cited and referenced materials to reflect the actual similarity index in context.
- **Without Reference:** The report excludes references and citations to assess the core originality of the submitted content.

The similarity index is within the acceptable limit prescribed by the university's academic policies, indicating that the report maintains academic integrity.

With Reference

B.Tech Team#10 Project Report

ORIGINALITY REPORT

11%	8%	11%	6%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

Without Reference

B.Tech Team#10 Project Report

ORIGINALITY REPORT

1%	0%	1%	0%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

Contribution to PSOs & Sustainable Development Goals

Title of the Project: Project Title
Project Members Name & Register Numbers: Student 1 Name (CB.EN.U4CYS220XX), Student 2 Name (CB.EN.U4CYS220XX), Student 3 Name (CB.EN.U4CYS220XX), Student 4 Name (CB.EN.U4CYS220XX)
Guide Name & Designation: Dr./Mr. GuideName, Guide Designation.

Contribution to PSOs:

(The below is an example. Please add based on contributions from your projects to the PSO. Remove this line after adding.)

- **PSO 1:** The project demonstrates a comprehensive understanding of emerging cyber threats such as clipboard hijacking, unauthorized data access, DLL injection, and replay attacks. By introducing innovations such as frequency-based clipboard monitoring, application authentication via DLL injection, and replay resistance mechanisms, the project reflects a deep awareness of the evolving cyber security landscape and an ability to anticipate and mitigate future threats. In addition, the project improves the confidence of stakeholders through a secure design and robust justification of security features.

- **PSO 2:** The project makes effective use of standard tools and technologies, including Windows API functions (SendMessage(), WM_GETTEXT, WM_CLIPBOARDUPDATE), Detours API for function hooking, and WebSocket Secure (wss://) for encrypted communication. The integration of encryption protocols, secure clipboard dispatch mechanisms, and a browser extension further showcase the application of modern cyber security practices. By ensuring privacy-aware data handling without compromising system performance, the project exemplifies the ability to design and

implement innovative cyber security solutions using best practices and industry-relevant tools.

Contribution to SDGs:

(The below SDGs are examples. Please add SDG based on the contributions from your projects. Remove this line after adding.)

SDG	Contribution
3. Good Health and Well-being	Enhances cybersecurity in healthcare systems to protect patient data and ensure availability of critical systems.
9. Industry, Innovation and Infrastructure	Promotes resilient infrastructure and fosters innovation in secure digital services.
16. Peace, Justice and Strong Institutions	Supports the development of secure systems that reduce cybercrime and build trust in institutions.

SDG 3

SDG 9

SDG 16